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1. The Big Picture

Nature has populated dark matter halos across a huge range of masses from the most massive giant ellipticals ($M_\star \sim 10^{12-13} M_\odot$) to grand design spirals ($M_\star \sim 10^{10-11} M_\odot$) like our own Milky Way, down to the smallest things we call “galaxies”: the so called “ultra-faints” ($M_\star \lesssim 10^5 M_\odot$) that are now known to populate the halo of our own Milky Way. These dwarf galaxies are, by virtue of their small size and (relative) simplicity, exquisite indicators of how the processes of gas accretion, star formation, energetic and radiative feedback, and interactions drive galaxy evolution. Further up the mass scale, we find spheroidals that have lost their gas and ceased to form stars long ago, and irregulars that have continued to form stars even while remaining low in metallicity. Yet, our present samples of these smallest galaxies have been limited to a narrow range of “host” environments that might obscure novel properties in a more diverse cosmic sampling of dwarfs. NASA’s next-generation flagships (*Webb* and *Roman*) will push the “edge of dwarf galaxy studies” well beyond the halo of the Milky Way, deep into the “field” and into the halos of other diverse hosts in our Galactic neighborhood.

We propose a new program that will deliver a novel suite of cosmological simulations and derived mock datasets optimized to study the smallest galaxies, as a rigorous function of their home environment, as they will be examined by NASA’s current and future flagship observatories. Our simulations are motivated by three specific science questions, all drawn from the stated goals of the forthcoming observations:

- Q1:** How do the smallest and faintest galaxies acquire, process, retain, and/or lose their gas to populate their dark matter halos with stars?
- Q2:** How do the faintest galaxies produce, eject, retain, and/or recycle their metals?
- Q3:** What role does *environment* play in all these processes?

We will leverage the cosmological hydrodynamic Enzo simulation code with new physics modules specifically designed to break new ground in dwarf galaxy resolution while also fully accounting for the influences of their environments. New code modules for achieving high mass resolution in dark matter, gas, and stars will enable us to meet or exceed the current limits for dwarf galaxy resolution while also explicitly tracking gas accretion, star formation, feedback, and chemical enrichment in the dwarf population as a function of environment. This combination of resolution and environment is the distinguishing feature of our proposal with respect to previous work, motivated by the need to capture the small-scale structure of diffuse gas at very fine *mass* resolution in and around statistically significant populations of galaxies.

Over the course of this program we will (1) extract a new set of “zoomed” and “multi-refined” dwarf galaxy simulations from an existing large-box precursor simulation, (2) run highly resolved simulations of a representative sampling of dwarf galaxies over a range of mass, (3) translate these simulations into a set of simulated *Roman* and *Webb* data products using an upgraded mock data pipeline, and (4) release public data products for use by other astronomers in the planning and interpretation of their observational programs.

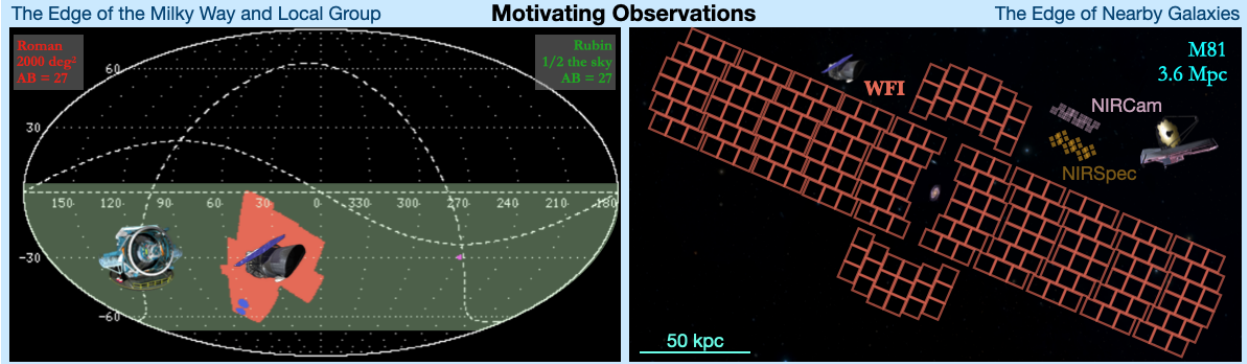


Figure 1 : Our program is motivated by NASA’s *Roman* and *Webb* observatories, complemented by the Vera Rubin Observatory survey. *Roman* will map faint dwarf galaxies out to beyond the edge of the Local Group (left), inside the Rubin footprint. With its large WFI FOV, *Roman* will also map the halos of nearby galaxies over large areas, yielding highly complete surveys of dwarf galaxies to large distances (right, a hypothetical footprint for M81). *Webb* will push NIR studies of dwarf galaxy star-formation histories and chemical enrichment to new limits using its exquisitely sensitive imaging and low-resolution spectroscopic modes, which can be applied in the Milky Way halo, the Local Group outskirts and nearby field, and in external halos (as shown for M81).

2. Relevance to NASA

Our goals derive from NASA’s strategic Objective 1.6 to “Discover how the Universe works. . .” and the ‘Cosmic Origins’ theme of SMD’s 2020 Vision for Scientific Excellence: ‘How does the universe work? How did we get here? Are we alone?’ Our simulations are *specifically motivated* by particular observations that will be made by NASA’s next-generation NIR flagships - the *James Webb Space Telescope* (now operating) and the *Nancy Grace Roman Space Telescope* (2027). Both of these missions will open new discovery spaces, and we have tailored our simulation plan to align with the key datasets. *Webb* can reach faint stars in the Milky Way halo and Local Group (LG) outskirts, measuring star formation histories and initial mass functions for ever more distant populations (e.g. DD-ERS Program 1334¹, GO Programs 1617² and 3849³.) We also expect that *Webb* slitless spectroscopy will be able to obtain metallicity estimates for some faint dwarfs out to the LG edge. *Roman* will open an entirely new discovery space for Galactic halo substructures, with its large area survey of the high latitude sky and its efficient coverage of nearby galaxy halos (Figure 1). In particular, *Roman* is expected to reveal (potentially) hundreds of new ultra-faint dwarf (UFD) galaxies in environments far away from the Milky Way where the limit of galaxy formation is unknown, and to extend the characterization of faint dwarfs to the halos of nearby galaxies. Both environments are essentially *terra incognita* for the faintest dwarfs. In addition to novel simulations, we will provide physical models and synthetic data for comparisons to observations by *Webb*, *Roman*, and the upcoming Rubin Observatory. With *Webb* already operating and with *Roman* well on its way in development, it is important that we begin this work now.

¹<https://www.stsci.edu/jwst/science-execution/approved-programs/dd-ers/program-1334>

²<https://www.stsci.edu/jwst/science-execution/program-information?id=1617>

³<https://www.stsci.edu/jwst/science-execution/program-information?id=3849>

3. Roman, Webb, Rubin, and the Edge of Galaxy Formation

3.1. New Discovery Spaces

The forthcoming observations will push into new frontiers in dwarf galaxy discovery. Figure 2 shows the locations of known dwarfs as a function of their V-band magnitude and distance from the Milky Way. The points mark the currently known dwarf population. Moving from the lower right to the upper left in this diagram means moving to fainter dwarfs that are farther away, which introduces a strong luminosity bias against the faintest dwarfs – they are simply too faint or contain too few stars brighter than the imaging magnitude limit to be detected as distinct objects. The more distant dwarfs are bounded, more or less, by the limits of the Sloan Digital Sky Survey (SDSS), a 2.4 m telescope that imaged a quarter of the sky to $AB \sim 22$. Close to the Milky Way, there are fainter known dwarfs, but all have been found in “pencil beam” surveys and are therefore an incomplete sample.

Enter *Roman*, *Webb*, and Rubin. *Roman*’s High Latitude Survey (HLS) will reach ~ 5 mag deeper than SDSS, pushing the limits for dwarfs at $M_V \sim -4$ out to 2 Mpc, and dwarfs at $M_V = 0$ (if there are any!) out to the Milky Way’s virial radius ($\simeq 300$ kpc). Rubin’s 10-year co-add imagery (which will overlap the HLS) will go to about the same depth with complementary optical photometry. The environment of dwarf discovery will soon extend to > 1 Mpc away from the Milky Way, and into the field. Until now we have not had the observations to intensively study the behavior of the faintest dwarf galaxies in these regions. *Webb*’s deep follow-up imaging will provide CMD-based star formation histories and perhaps even metallicity distributions from low-res spectroscopy of faint galaxies that *Roman* and Rubin discover. This is unexplored territory.

This new discovery space motivates our choice of “the edge of galaxy evolution” as the science theme of our program, because it will soon be possible to reach the outside

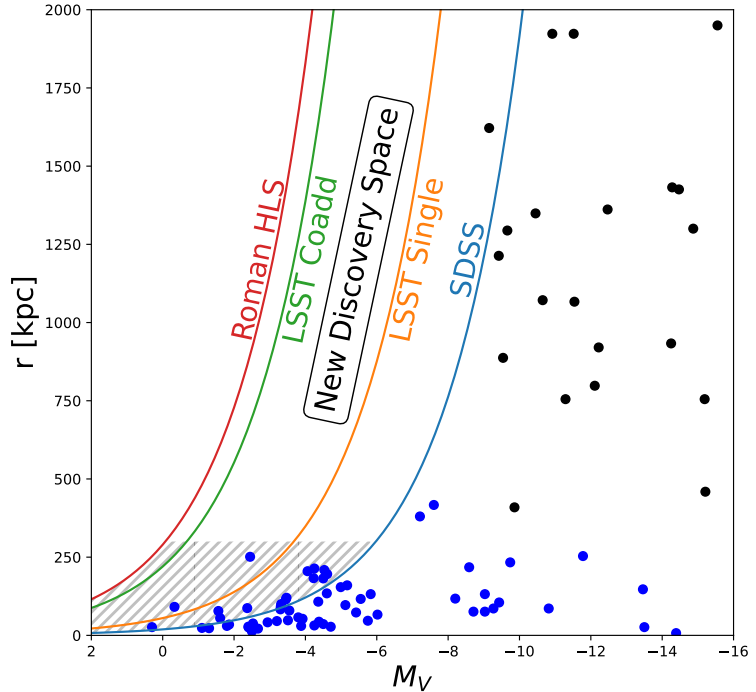


Figure 2 New discovery space for faint dwarf galaxies, based on [50] with updates for *Roman* HLS and the Rubin/LSST large area surveys. The small circles mark known dwarfs in the Milky Way halo (blue) and Local Group (black) from recent review articles [29, 45]. The grey shaded region between the SDSS and HLS curves marks the large new discovery space where *Roman*, Rubin, and *Webb* will search for new populations at the edge of galaxy evolution. Note that we have omitted dwarf satellites of Andromeda due to the very different detection systematics there relative to the rest of the Local Group.

edge of our Local Group and the faint edge of occupied halos. *Roman's* deep imaging will cover the halos of nearby galaxies out to hundreds of kpc (Figure 1), yielding far more complete samples of dwarfs than ever before. *These new discovery spaces raise the exciting possibility that we can understand how all the processes that govern dwarf galaxy evolution, especially at the faintest end, depend on the environment in which these dwarfs find themselves over the course of their evolution.* Environment may prove critical: there are already strong indications that dwarf galaxies outside larger host halos are more gas rich [17] and may have lower metallicities [24], that the satellites of passive, quenched galaxies are more quenched themselves (conformal quenching) [54], and that Local Group analogs in the nearby universe have a satellite diversity that correlates with environment [27]. Moreover, recent observations have pushed studies of satellites near the UFD regime to the wider universe [27, 53, 9, 4]. While not quite able to reach the edge of the UFD population, these observations reveal that the Local Group dwarf galaxy population is but one example of such systems, and other comparable host galaxies may host quite different populations. So while environment may not be everything, it may be a very large factor in how dwarf galaxies evolve.

Our top level science goal is to understand how dwarf galaxies across a representative range of mass and environment acquire their gas, form stars, eject feedback, distribute metals, and cease forming stars (or not), under the influence of their internal processes and their environments. This explicit tracking of environment for high-resolution dwarf galaxy simulations is a new, open field for theory that we will explore with a set of novel techniques optimized for this purpose.

3.2. Pushing Back the Edge of Galaxy Evolution - A Brief Review

The Sloan Digital Sky Survey opened a new frontier in our understanding of galaxy formation at what might be the “bottom end” of the process. Beginning in 2005, the large areas of the SDSS were mined for small, clustered stellar populations that soon became known as the “ultra-faint” dwarfs [61, 62, 46]. Over the following 18 years, intensive studies of this population have characterized their overall abundance in the Milky Way halo [60], their metallicities and chemical abundances [31, 28, 46, 22], and their internal kinematics [23, 31] and from this, inferred the properties of their dark matter halos [48, 63].

The discovery of the ultra-faints soon led to the suggestion that they are “fossil galaxies”, possessing only old stellar populations that were formed before or during cosmic reionization and unable to retain gas and form stars since that epoch [39, 5, 42]. Deep imaging with *Hubble* has largely borne out the relative absence of intermediate or young stellar populations in the UFDs, which show star formation histories (derived from extremely precise color magnitude diagrams) that are consistent with all of their stars being formed prior to the $z \sim 6$ end of cosmic reionization [6], in contrast to the classical dwarfs known before SDSS that have more diverse and extended star formation histories [55, 10]. The discovery of the UFDs also quickly sparked a debate about whether they could explain the so-called “missing satellites”—the large discrepancy between the theoretical mass spectrum of CDM halos and the known populations of galaxy subhalos—that now seems to have settled on the conclusion that there is no large discrepancy remaining, once a number of selection biases are taken into account [50, 21].

Because of current observational surface brightness limits (Figure 2, nearly all known

UFDs are close to the Milky Way, which may bias our understanding of them. Indeed, recent *Hubble* observations of UFDs in and around the M31 halo are challenging the idea that all UFDs are fossil galaxies [30, 44]. *Roman* and Rubin will enable detection and characterization of UFDs not only around more distant hosts, but also far into the field, where their evolution has not been heavily influenced by more massive neighbors. This new observational parameter space demands a similar leap in the simulations of dwarf galaxies that explicitly track environment as a key influence.

3.3. Simulated Ultra-faints So Far

We will focus on the UFDs for the purpose of this present discussion, because the UFDs generally present the most compelling open questions (to us, anyway) and set the most demanding technical requirements for simulations. While we use UFDs to motivate and bound the discussion, we intend to simulate and study a wide mass range of dwarf galaxies ($M_{\text{DM}} \sim 10^8\text{--}10^{10}$) in diverse environments to meet our goals.

In parallel with the observational advances outlined above, UFDs have attracted interest from simulators who have addressed many of the open questions about the known Milky Way population. Before introducing our simulation plan (§4), we survey here the existing literature on UFD simulations. The current state of the art breaks down into two major categories of simulation: those that track UFD halos within the halos of larger galaxies (usually $\sim L^*$ hosts) and those that study “isolated” dwarfs separated from any surrounding environment, so that computational resources can be focused on the UFD itself.

From 2015–2023, a number of groups have simulated UFDs in isolation or in groups with other dwarfs; generally these UFDs were selected for dark halo masses of $\sim 10^{8\text{--}9} M_{\odot}$. The resolution of these isolated UFD simulations is generally very high because they have been optimized to spend computational resources on the UFDs themselves [35, 57] rather than on L^* hosts or larger environments. These simulations generally have dark matter particle masses of one to a few thousand $M_{\odot}$, down to $\sim 100 M_{\odot}$ at the current state of the art [59, 1, 18]. These latter simulations are tours-de-force of the zoom-in art, with star particle masses of 20–30 $M_{\odot}$ and effective spatial resolution (e.g., the smoothing length) of ~ 10 pc. These studies have, variously, found that an early cosmic rise of radiation in combination with supernova feedback can truncate star formation [35, 26] but that star formation can be rejuvenated in some circumstances [64, 38]. The mass assembly pathway and merger histories also appear to matter in setting both the stellar-mass-to-dark-matter-mass ratio and its scatter [26, 37]. Because of the diversity of merger histories, it may be that even these “simplest” of galaxies are in fact composite stellar populations formed apart and only seen as one coherent cluster much later [20]. The downside of the tradeoff for extreme resolution is that these simulations really are isolated, with no context, making it impossible to study environmental influences in a controlled manner.

These “isolated” UFD studies are the best attempts to characterize and understand the internal evolution of these small galaxies, and they have sifted through many of the key physical processes to identify those that matter most. However, there are still major open questions and uncertainties related to both physics and numerics. We will focus on the well-known persistent questions: the stellar mass / halo mass ratios (related to the luminosity function) and the mass-metallicity relation. These are two yet-to-be-understood patterns that will be addressed by future observational data.

3.4. Capturing Gas Retention and Loss: The Key to the Luminosity Function

Whatever *Roman*, *Webb*, and Rubin find in that blank spot on the map in Figure 2, understanding the relationship between stellar mass and dark halo mass⁴ is the critical puzzle piece: without stars, these little subhalos will not be seen. This prompts perhaps the most critical question: how many halos form stars, and how many stars do they form? When do they start and why do they stop? Existing simulations show that even very small mass halos can be occupied with stars, down to the $M_{\text{DM}} \sim 10^7\text{--}8 M_{\odot}$ range [3, 33, 16]. Yet even (or perhaps especially) in these small halos, determining how and when they retain gas and form stars is not a simple matter, as there are many possible influences. It is already well established, through the concept of the “filtering mass” [15, 14], that external radiation from the “metagalactic background” that ramps up at reionization can ionize the gas in and surrounding small halos, making it difficult for them to retain cold star forming gas in their shallow potential wells. This effect is generally invoked to explain the fact that the UFDs as a population are consistent with having only very old (> 10 Gyr) stellar populations [6]. As we go up the mass scale of Milky Way/Local Group dwarfs more extended star formation histories start to appear, suggesting that some halos are able to retain gas and form stars over longer periods [56]. Yet, as we have argued above, these conclusions are based on samples that are highly biased towards the Milky Way halo (inside R_{vir}), which is not a typical cosmological environment today and certainly not representative of the environment in which most dwarf galaxies form over cosmic time.

We are thus left with open questions: What processes or balance of processes, and/or environmental effects actually set the gas supply? What do the smallest dark matter halos do outside of L^* environments? Can they continue forming stars longer than if they are subhalos? Can they experience a “rejuvenation” of star formation as some of their more massive counterparts have done inside the Milky Way halo? These questions compel an approach that can track not only the loss of a small halo’s gas supply, but also other possible environmental influences. These include ambient gas present in the large scale structure, mergers and interactions, and the circumgalactic medium of hosts. To understand these issues, we must capture the evolving gas supply of small halos AND their proper environmental context in the same self-consistent simulation. We will address these questions with new simulations that specifically track faint dwarf galaxies as a function of environment and that focus resolution on the physical processes that govern the gas supply. This set of questions will form the basis of analysis and papers to be led by Institution A (PI A1 and Co-I A2) once the simulations are complete (technical details in §4–5).

3.5. Capturing Metal Mixing: The Key to the Mass-Metallicity Relation

The isolated dwarf simulations described above have generally learned much, but still struggled to match the mass-metallicity (MZ) relation for UFDs owing to a combination of uncertain physics and limited resolution [1]. This is unfortunate, since the placement of the UFDs on the *observed* mass-metallicity relation is one of the most robust and well-behaved observational results for UFDs so far [22]. Reaching an understanding of how

⁴This is sometimes called the “SMHM” for “stellar mass to halo mass” relation. We prefer to use “mass-to-light” or “ML” ratio to avoid this jargon.

this clear relation is produced would be a significant advance in understanding how chemical evolution works at the smallest mass scales, and by extension, at the beginning of cosmic evolution.

Our working hypothesis is that the ejection, mixing, and/or recycling of metals is a critical process in setting the UFD mass-metallicity relation, and that a lack of resolution in the gas that receives the products of supernovae may be a factor inhibiting the successful recovery of the observed MZ relation. A simple calculation highlights the challenge of handling metals in numerical simulations: if, say, the $0.1 M_{\odot}$ Iron production of *one* core-collapse supernova is ejected from a nascent UFD, the magnitude of the gas resolution element imposes limits on how extensively this “yield” can mix. If the fluid resolution element is much larger than this—say, hundreds to thousands of solar masses—then metal mixing will be artificially effective, high metallicity gas will be artificially suppressed, and even if the observed MZ relation is matched, it could be for the “wrong” reasons. This line of reasoning about the metal mixing process leads directly to our requirement for simulations with smaller gas resolution elements than have been achieved to date.

Metals are Nature’s tracer particles, giving unambiguous evidence that stars have influenced the gas where they are found. The mass-metallicity relation for dwarfs is well-behaved inside the Local Group, but of course this is a biased sample, and even this result has proven hard to recover in simulations. The open questions are: How do dwarfs across the mass range create, eject, retain, and/or recycle their metals? How and when do galaxies with such simple star formation histories exhibit relatively wide spreads in metallicity? Do dwarf galaxies in larger halos share metals with the host and do dwarfs in low-density (field) environments have simpler chemical enrichment histories? What

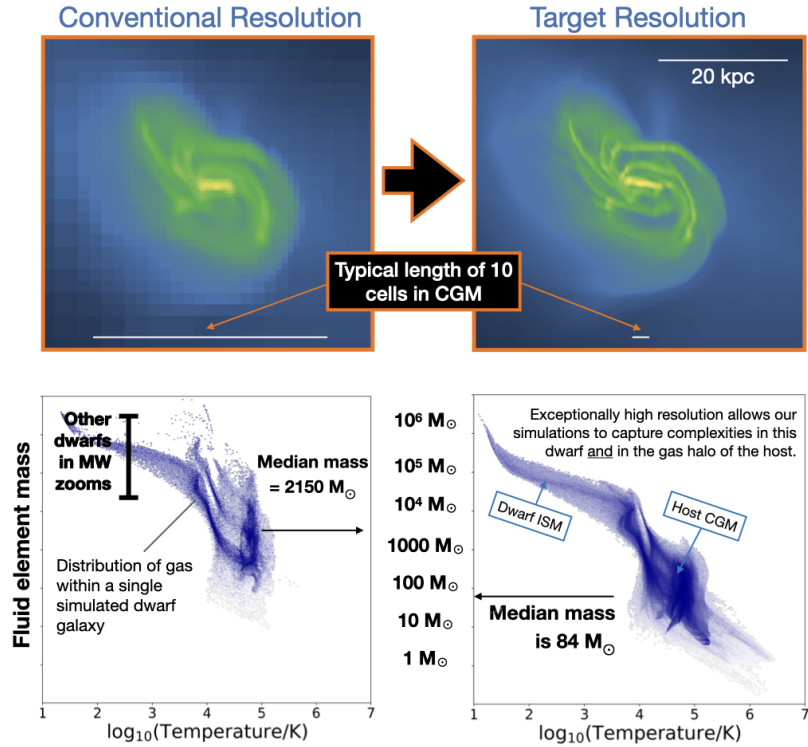


Figure 3 : An illustration of the gas resolution for a precursor dwarf simulation. Left panels show a conventional ‘zoom’ simulation in which fluid elements are refined based on density. We distinguish studies with UFDs evolving in L^* halos [64, 2, 3, 33, 16] and those evolving in zoom simulations within field environments [35, 57, 58, 26, 20, 32, 37, 59, 1]. Our adaptive mesh refinement (AMR) approach (right panels) naturally reaches very high gas resolution because the fluid elements are set by adaptive cell size rather than mass. *Our planned simulations will reach at least $2\times$ smaller cell sizes and perhaps $8\times$ smaller masses than this prototype.*

role does interaction with the large scale gas environment play in distributing metals? Our simulation plan will apply gas resolution beyond the current state of the art to explore how metals are distributed and mixed by and within the dwarfs and exchanged with their environments. As with the star formation histories and stellar masses, we must capture the evolving gas AND metal supply of small halos in their proper environmental context in the same self-consistent simulation. This set of questions will form the basis of analysis and papers to be led by Institution B (Co-I B4 and an undergraduate student) once the simulations are complete (technical details in §4–5).

3.6. From Science Goals to Simulation Requirements

After considering the state of the art and the open questions in dwarf galaxy evolution, we find that there is a significant opportunity for new simulations. **We will approach key problems from a new direction: by increasing the resolution in the gas and tracking its evolution under environmental influences.** The cosmic abundance and spatial distribution of faint dwarfs that occupy *Roman* and *Webb*’s new discovery space (Figure 2) depend directly on how (or whether) they have been populated by stars. These stellar masses depend on how long the dwarfs can retain gas and form stars, a process we already know to be stochastic and probably subject to environmental influences. Gas retention in turn depends on how energetic feedback, the *external* gas environment, and ambient radiation influence their gas supply. The mass-metallicity relation also depends on how gas is retained (as the reservoir for diluting metals) but also on how metals are mixed and/or recycled in these shallow potential wells. Mitigating over-mixing of metals is a particularly important requirement, as real dwarfs approach a truly stochastic regime of chemical enrichment at smaller masses and numerical over-mixing can erase this effect.

Figure 3 shows what is possible. The colormap plots a distribution of fluid element masses as a function of temperature for a test precursor dwarf simulation. This dwarf happens to be in the halo of an L^* host, but need not be. With AMR simulations, we can reach fluid mass elements well below $100 M_\odot$ in the gas surrounding dwarfs, and even down below $1 M_\odot$ in the hot gas that carries supernova ejecta out to be mixed with cold gas. The vertical black line shows the approximate range of masses for single gas particles UFD simulations from the literature. The gas resolution we expect to achieve in the production simulations is $8\text{--}10\times$ better than this prototype, and will be deployed on a large representative sample of dwarfs—not just one. This ability to reach new gas resolution limits offers a route to understanding dwarf galaxy physics in a gas-focused way that complements existing work and yet is aimed directly at the expected datasets.

From this point of view, the big picture of dwarf galaxy discovery in the next decade draws a direct line from observational expectations to a computational approach that resolves the gas as much as is practical. We will now turn to a description of our plans for simulations that meet these requirements.

4. The Simulation Suite

Our goal is to understand the ways that environmental history affects the evolution of dwarf galaxies via large-scale cosmological adaptive mesh simulations using the Enzo code (Sec. 5). Our simulation plan is thus crafted with this aim in mind: starting by determining the large-scale environments of dwarf galaxies, and studying them in a range

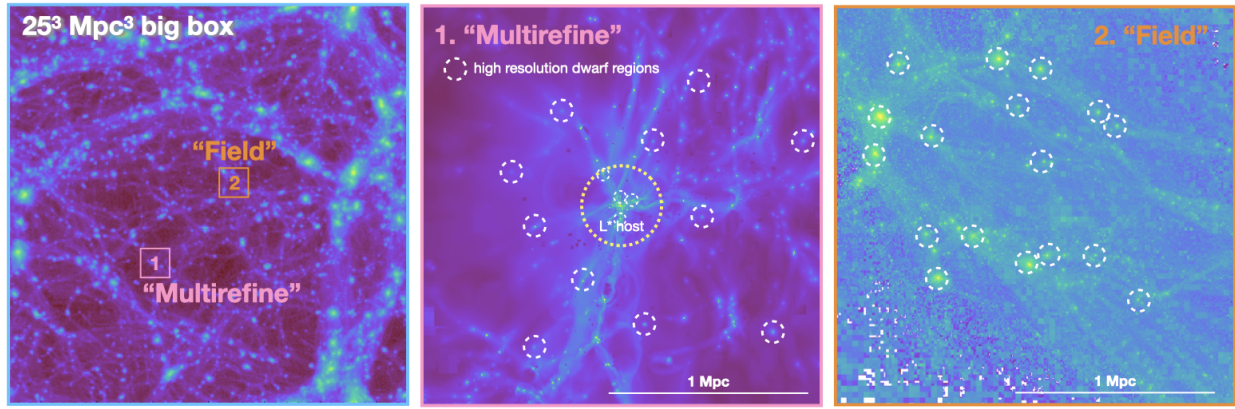


Figure 4 : We have already run prototype simulations for all program elements. The parent domain (left) is available from a separately-funded project. We will add the “multi-refine” method in and around L^* halos (middle) and the “field zooms” of low-density regions (right). These simulations are real but the figure only illustrates concepts, not exact locations or number of zooms.

of environments using two kinds of zoom simulations: (1) “multi-refine” simulations, wherein we add extra resolution to dwarf galaxies within $6R_{\text{vir}}$ of L^* galaxies (~ 1.5 – 2 Mpc, as motivated by Figure 2) in order to study both dwarf satellites and those “at the edge,” and (2) low-overdensity volumes consisting solely of field dwarf galaxies. Our aim is to produce ~ 5 of each kind of simulation to account for cosmic variance, but even a single calculation of each type will give us dozens of dwarf galaxies spanning decades in stellar mass ($M_* \sim 10^{3.5} - 10^{8.5} M_\odot$) in a wide range of environments, providing a new window into how environment shapes dwarf galaxies through cosmic time (see Fig. 5).

4.1. Selecting Environments from the Large-Scale Structure

All of the “zoom-in” cosmological simulations will be selected from the same $(25/h)^3$ Mpc³ volume (the “parent big box”), which has already been run as part of another project. This will enable us to understand the environments and histories of these halos and subvolumes in a full cosmological context with high integrity. This allows us to select for further refinement based on environmental criteria and so crucially enables our study of the effects of the environment on dwarf evolution.

4.2. Dwarf Galaxies Around L^* Halos

The key discovery space opened by *Roman* (Figure 2) is the far outskirts of the Local Group and nearby field, so our simulations must be able to track dwarfs with high resolution while explicitly tracking their environments. This kind of simulation is illustrated in the top-right panel of Figure 4. We will start with the now-typical “zoom-in” simulation of L^* galaxies that have extra refinement added to the gas surrounding the galaxies (the CGM, [36, 47, 52, 49]). Our runs will impose a resolution “floor” ensuring that the ambient gas felt by dwarf galaxies is always refined to a minimum level in the CGM, as well as a cooling refinement scheme that allows gas with a cooling length shorter than the floor resolution to reach smaller length scales where it needs to. These CGM resolution levels will follow a nested-refinement technique wherein we relax the refinement criteria farther from the host galaxy, starting with a floor resolution of ~ 100 pc within R_{vir} and trailing off to ~ 1 kpc at $\gtrsim 6R_{\text{vir}}$ (roughly 1.5–2 Mpc). Resolving the CGM and IGM gas

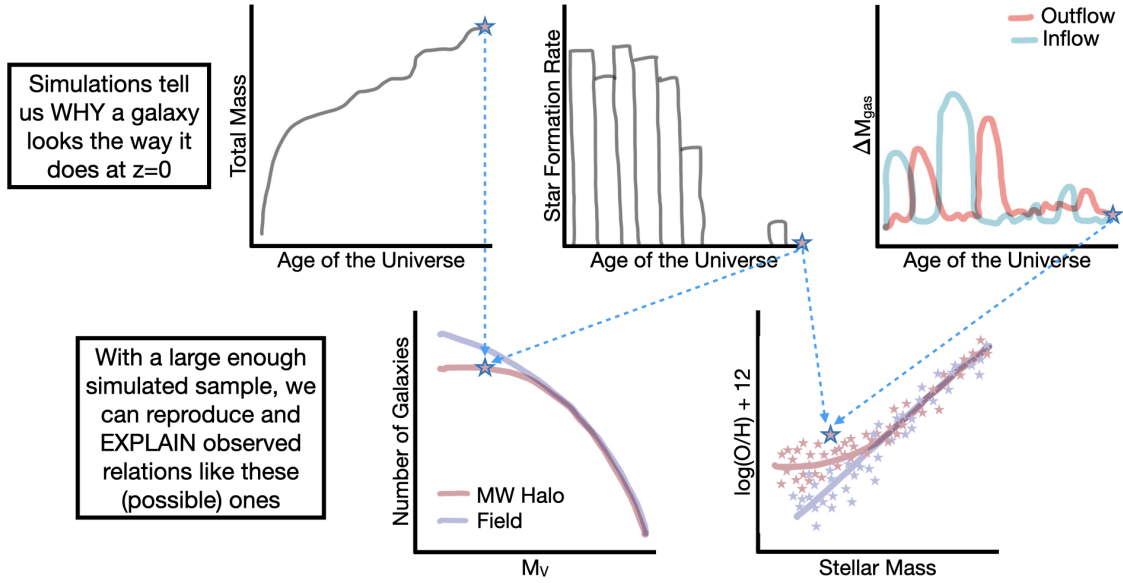


Figure 5 A schematic cartoon for how we can use theoretical calculations to understand observable quantities. For each dwarf in the sample, we will have ‘evolutionary tracks’ that follow its dark matter mass (M_{DM}), star formation rate, stellar mass (M_*), the density and temperature of the changing local environment, etc. These tracks, sampled at gas and stellar mass increments as small as $\sim 100 M_\odot$, will explain why each dwarf ends up where it does on the observed relations in the lower row. Studying a sample in which each dwarf is analyzed in detail this way will provide rigorous physical explanations of the observed relations that might otherwise be attributed to mere stochastic variation.

is *crucial* to following how environment shapes dwarf galaxies [e.g., 47]—to understand how filaments, outflows from larger galaxies, and other environmental phenomena affect dwarf galaxies, we must first resolve these diffuse features.

Within these L^* nested refinement zooms we will apply a new “multi-refine” technique (§5.2.3) that will impose high resolution simultaneously on *multiple* dwarf galaxies surrounding the more massive host (see Figure 4). The ISM of these galaxies will be allowed to reach comoving spatial resolutions of $\lesssim 30$ pc with their individual CGMs reaching spatial resolutions of 50–200 pc in nested refinement spheres out to ~ 25 –50 kpc (i.e., ~ 1 – $2R_{\text{vir}}$ for a $10^9 M_\odot$ halo), where it will meet up with the ambient gas resolution of the host described above. We will track hundreds of dwarf galaxies over a wide range of halo masses (and thus expected stellar masses) to include at least roughly a few dozen galaxies per dex in mass over the full suite of simulations. Owing to the availability of the parent big box and selected L^* zooms inside it from a separately funded program, we can pre-select L^* central galaxies and some satellites efficiently. We will target dwarfs with stellar masses between $10^{3.5} M_\odot$ and $10^{8.5} M_\odot$, with a particular focus on how gas and metals are retained or lost in galaxies with $M_* < 10^6 M_\odot$. Within these parameters we will also aim to select dwarfs with a wide variety of orbits, infall times, and present-day distances from their hosts in order to better sample the range of dwarfs observed around real MW-mass galaxies. The new runs of these L^* galaxies for this program will add the multi-refined dwarf ISM/CGM resolution as well as better stellar mass resolution. Dark

matter mass will be $\simeq 2 \times 10^3 - 2 \times 10^4 M_\odot$, and thus we expect to capture satellites down to $\simeq 10^7 - 8 M_\odot$ in total mass, covering the UFD range.

Our goal is to simulate the dwarf satellite populations of five $\sim L^*$ hosts that sample different environments and formation histories, sampling a total of 250–500 highly-resolved dwarf galaxies in a large range of realistic environments. In addition to examining the evolution of these galaxies, these calculations will provide valuable information about the formation of the stellar halos of L^* galaxies (including stellar streams, also a science use case for *Roman*). These simulations are targeted specifically at Milky Way halo observations (with mock data viewing the simulations “from the inside” [e.g., 65, 43]) and also to the wide-field samples of dwarfs around external galaxies expected from *Roman*.

4.3. Dwarf Galaxies In The Field

Figure 4 illustrates our simulation scenario for field dwarf galaxies. We will select 5-10 low-overdensity regions of radius $\sim 0.5 - 1$ Mpc from the parent big box that contain no halos more massive than $10^{9.5} M_\odot$ at $z = 0$ – i.e., that contain only dwarf galaxies that are not directly influenced by more massive neighbors – so that we can get a measure of dwarf behavior in the “true field”. The prototype in Figure 4 has ~ 20 dwarf galaxies at $M_{\text{dm}} \gtrsim 10^8 M_\odot$, and will resolve many more at the higher target resolution. The production runs should be able to apply our highest resolution to at least this many galaxies in a given zoom region. Simulating many dwarfs at once will capture a range of small-scale environments, interactions, and diverse merger histories, all of which appear to matter as influences. We will re-simulate these regions as effectively large “zooms”, with high dark matter resolution ($2 \times 10^3 - 10^4 M_\odot$) and our highest ISM/CGM/IGM spatial resolution (~ 30 pc), using the same cooling, star formation, and feedback physics as in the simulations described in Section 4.2. The $\sim 100 - 200$ dwarfs in the field regions will achieve spatial resolutions at least as good as in the L^* multi-refinement runs, and the prototype run shown suggests there is a chance that these low-density regions will be computationally efficient enough that we can increase their spatial resolution still more for one or more runs (to $\lesssim 15$ pc), which would be competitive with the highest resolution simulations of isolated dwarfs so far. These simulations are targeted specifically at observations of dwarfs far away from the Milky Way, Local Group, and from external galaxies—those objects in the blank upper reaches of Figure 2 where *Roman* and Rubin will fill in the map.

5. The Tools for the Job

In this project we will use the Enzo code [8, 7]. Enzo is a publicly available Cartesian adaptive mesh refinement (AMR) code used for the simulation of cosmological and astrophysical phenomena. It uses a block-structured adaptive mesh refinement scheme to achieve high spatial and temporal resolution, and combines an N-body adaptive particle-mesh solver for dark matter dynamics with a Piecewise Parabolic Method (PPM) hydro solver that has been extensively modified for cosmological applications and hypersonic flows. In addition, the code includes a range of other physics including both equilibrium and non-equilibrium chemistry and cooling models, radiative cooling using a Raymond-Smith model and Cloudy emissivity tables, and prescriptions for the formation and feedback of both stellar populations and black holes. Enzo also includes modules for magnetohydrodynamics (both the Dedner divergence-cleaning method and a constrained trans-

port method) and radiation transport using a ray-casting method and flux-limited diffusion. Enzo has been used to model a wide variety of cosmological and astrophysical phenomena – most relevant to this proposal, Enzo has been used extensively to study high- and low-redshift cosmological structure formation, the intergalactic and circumgalactic medium, and the properties of star-forming molecular clouds.

5.1. Key Advantages of the AMR Approach

AMR has significant advantages that complement the best attributes of the SPH and moving mesh techniques. AMR utilizes a grid with set physical cell sizes which are adaptively refined according to specified criteria, so it can, e.g., refine where the density or metallicity is high, or where cooling time is short. Within these cells the gas *mass* can be highly variable depending on local conditions (see Figure 3 for a prototype). This generally contrasts with particle or moving mesh codes, which typically have a fixed fluid element mass but a varying space occupied by that element, so that “mass resolution” is effectively constant within a given domain. Both methods have their advantages: for AMR the advantage is the ability to achieve extremely high mass resolution ($\sim 100 M_{\odot}$) in low-density gas while keeping its finest spatial resolution competitive. While AMR cannot solve every problem, this trade-off uniquely enables our goals for properly tracking the baryonic processes of energetic feedback from supernovae, tidal stripping from mergers and interactions, and all the other gas flows that feed galaxies.

5.2. Project-specific Enzo Improvements

5.2.1. *Small Star Particles*

Hydro simulations cannot yet simulate L^* galaxies with only one “star particle” per star formed, while still capturing the full environments in which stars and galaxies form. However, several groups have brought their methods close to the ultimate “one-particle-per-star” limit for dwarf galaxies in cosmological simulations [26, 59]. To achieve our goals, and in particular to appropriately simulate UFDs, we will need to implement one of these methods for forming and evolving very small star particles to “re-baseline” Enzo’s star formation and feedback algorithms. Fortunately, it is not necessary to resolve every star—low mass stars can still be aggregated together in the usual fashion and evolved collectively throughout the calculation. Rather, we will use a hybrid method similar to that employed in other Enzo simulations [13] or recent SPH simulations [2] that tracks individual high mass stars with single particles and aggregates low mass stars together. The dividing line falls where the chemical yields become important for the key elements (α elements and Iron), i.e., $\simeq 8 M_{\odot}$ for Type II supernovae. We will use a prescriptive model for the time-delayed enrichment from double-degenerate Type Ia supernovae from the aggregated lower-mass stars, using a standard model [e.g., 25]. This method will enable explicit tracking of star formation and chemical enrichment (including, e.g., tracking metals formed from supernovae and AGB stars separately; § 5.2.2) in a way that is absolutely required at the dwarf mass scale we are pursuing. Given that the method has already been used and published for cosmological simulations, we are confident of its feasibility and consider this as a low-risk component of our program.

5.2.2. *Chemical Evolution*

Building on our treatment of individual massive stars, we will implement a flexible chemical evolution model that explicitly follows the astrophysical processes that enrich our simulated galaxies with heavy elements. This model will separately track nucleosynthetic enrichment from Type Ia and Type II supernovae, AGB stars, kilonovae and other potential r-process sites, and re-enrichment from stellar winds and planetary nebulae. It will also include a “total metallicity” field that will be used for calculating metal-dependent cooling. Following processes rather than individual species lends us flexibility in terms of making synthetic observations, as we are thus not strongly bound to any individual set of yields from stellar evolution calculations (which display substantial variation between model sets at low metallicities). In addition, we will ensure that energetic and nucleosynthetic feedback are self-consistently coupled to ensure the most accurate possible capture of the *distribution* of the nucleosynthetic products.

This model will be built on top of the open source JINA-NuGrid galactic chemical evolution pipeline [12], and in particular will leverage the SYGMA code [40] to produce extremely detailed and high resolution tables of nucleosynthetic and energetic yields from single-metallicity stellar populations as a function of time and nucleosynthetic source. This level of fidelity will be useful in two ways. First, it allows us to experiment with stellar initial mass function variations and different stellar evolution models (which can have different post-main sequence evolutionary behavior, particularly at low metallicity). Second, it provides us a platform for post-processing our simulations using detailed stellar yields to make predictions for spectroscopic followup with *HST*, *JWST*, and/or ground-based telescopes (which will be done by an undergraduate student at Institution B). Because this plan builds on proven tools with mild extensions, we regard this component as feasible and low-risk.

5.2.3. *“Multi-refine” Tracking of Dwarfs*

Explicit, high resolution simulations of representative populations of dwarfs is a key goal of our program. For all three of the “edge” scenarios—Milky Way outskirts, external galaxies, and the true field—it will be necessary to apply optimal resolution to the selected dwarfs while saving computational resources by NOT simulating regions far away and not of interest. To optimize the resolution for the selected dwarfs, we will implement, test, and then deploy a novel method for tracking multiple regions of interest within the same zoomed region – a technique we are calling “multi-refine” or “multi-zoom.” This is a generalization of techniques proven in other contexts [e.g., 52, 36, 19] that enforces refinement on particular regions that are either explicitly derived from previous lower resolution track-finding runs or algorithmically determined at runtime. For our project, we will combine pre-computed tracks with “tracer particles,” defined to be dark matter particles known to lie fully bound to a particular dwarf, which can then be used to locate a refinement box that follows a dwarf through the domain. These techniques have been implemented in Enzo for a single galaxy (which is really just an arbitrary contiguous volume), so we will spend part of the first year of our program implementing, testing, and deploying multi-refinement zooms on our platform. Figure 4 shows a cartoon of this method where dwarfs of interest are marked with circles for dwarfs in the outskirts of a MW-like halo (middle panel) or in a true field zoom (right panel). For the “halo” and

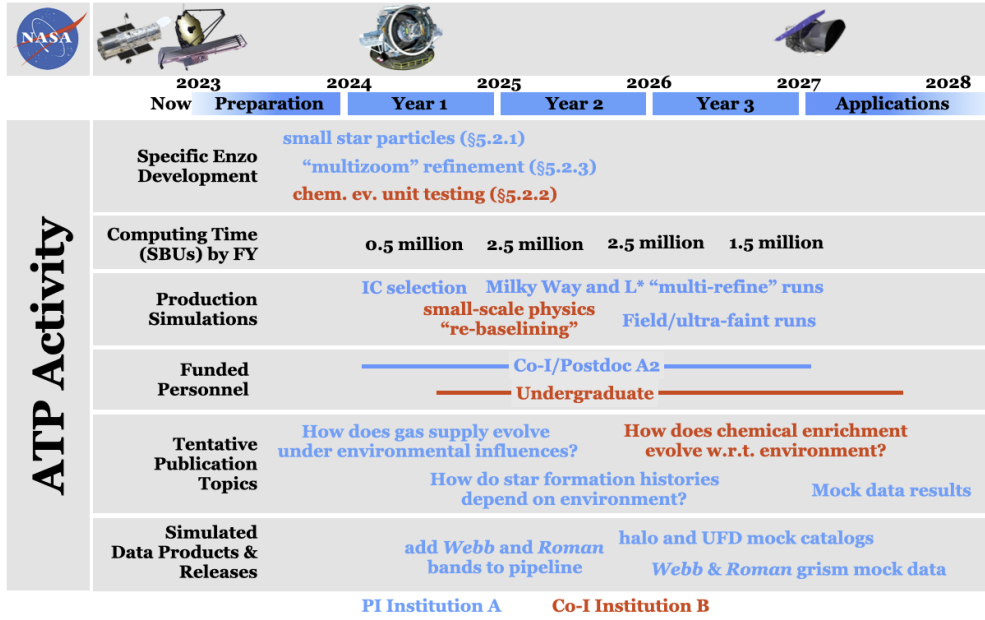


Figure 6 : A schematic integrated timeline of our major activities unfolding in parallel. The colors indicate the lead institution for that activity, even though others will usually be involved. This is designed to show the big picture of our efforts, not an exact schedule or work breakdown.

“outskirts” scenarios we will select samples of dwarfs that populate a range of parameters (e.g., local overdensity, orbit, assembly history), while for the “true field” environment it is likely that we will be able to follow all halos at and above the ultra-faint scale within each roughly spherical zoom region. Because it builds on proven, published techniques with only mild extensions, we regard this component as feasible and low-risk.

6. Synthetic Data: Products and Heritage

The goal of all astrophysical models is to predict and interpret observational results drawn from the real Universe. For our science goals, we need synthetic data in order to know which differences in physical processes will be discernible with what data—a catalog-based CMD, for example, is useful for discerning if observations can distinguish between stellar populations with different metallicities, but it is only with fully cosmological simulations backing those catalogs that we can bring the synthetic data back to the underlying physics of *why* two populations might have different metallicity distributions. We thus designed our mock data plans (Figure 7) with two important goals in mind: (1) to build dedicated pipelines for the specific observatories, instrument modes, and data properties of the observational programs driving progress in the next few years, and (2) to optimize resources (old and new) by leveraging existing tools to the greatest possible extent. These synthetic data (and associated physical information necessary for *using* the data) will be made publicly available as High Level Science Products (HLSPs) via STScI’s MAST.

The mock data products that are shown in Figure 7 build on an existing data pipeline developed and maintained by our group based on community tools (e.g., YT [51], Powderday [34], Hyperion [41], and FSPS [11]) that we will augment for the proposed program. The components to be added are *Webb*- and *Roman*-specific grism spectroscopy (from which dwarf galaxy metallicities can be inferred), mock color-magnitude diagrams for the *Webb*, *Roman*, and Rubin bands, and the intermediate products, such as star formation and metallicity histories, that provide the “ground truth” underlying the mocks.

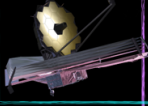
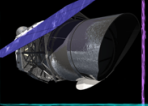
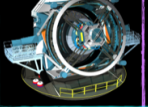
	Milky Way and Field Observations	Nearby Galaxy Halos and Dwarf Observations	Related Mock Data Products
	targeted Local Group UFDs star formation histories metallicity distributions initial mass function	targeted dwarf galaxy followup of Roman discoveries deep CMDs for star formation and age distributions	deep images to AB = 30 mag over ~3 arcmin fields grism spectra to AB = 26 mock CMDs in IR bands
	UFD statistics at > 300 kpc streams and stripped cores Programs: HLS, GO programs	“complete” dwarf populations star formation histories gas and dust content trends with host properties	wide images to AB = 27 deep images to AB = 29 mag grism data to AB = 25 mock CMDs in IR bands
	UFD statistics at > 300 kpc All-sky Streams Halo-disk interactions	large scale environment for hosts and dwarfs optical photometry for bright end of dwarf CMDs	dwarf and UFD mock catalogs to AB = 29 mag CMDs and metallicities in ugrizy bands

Figure 7 : This figure shows how the real datasets (in the first two columns) drive the requirements for our planned mock datasets (third column). The observations are broken down roughly into the “Milky Way and nearby field” and “External galaxies and nearby field” scenarios. The background is a synthetic image from a prototype L^* galaxy from our existing mock data pipeline.

7. Data Management Plan

Lower-level Data includes simulation outputs and metadata. Enzo creates outputs in HDF5 format using a custom self-encoding file format. Each data output in a time series will be between a few to 10s of GB, and we expect to create up to 20 TB of such outputs per run. Simulations evolved on Pleiades will have their outputs archived with the Lou Mass Storage System, a tape backup system that enables long-term preservation and access by interested scientists to the enormous (PB-scale) datasets that cannot be readily shared via the internet. This data will be made available to external users upon request; we do not currently have plans to make all of it public due to the technical challenges involved.

Software Infrastructure includes contributions to Enzo and the mock data pipeline in addition to code, scripts, and pipelines that operate on the lower level data to create higher-level data and prepare results for publications. Examples include:

1. We will share new Enzo modules (e.g., multi-zoom refinement, small star particles, chemical evolution enhancements) with the Enzo user community via regular pull requests.
2. Simulation analyses will build on, and give back to, the community YT code, via public GitHub repositories to develop, test, and share the analysis scripts and packages.
3. Our existing synthetic data pipeline will be extended as part of this project, and new modules will be open-sourced via GitHub.

High-level data are science-ready curated data products, including synthetic observations paired with select simulation data intended for re-use by both the team and the wider community. This will include halo catalogs, merger trees, partially processed projections and slices, pencil beams of extracted data, star formation histories, and halo data that has been extracted from the main simulation, all as a function of time. The smaller data products (e.g., § 6) will be made available via STScI’s MAST as HLSPs.

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9. Inclusion Plan

We recognize inclusion as a core NASA value that stands with and enables mission success. Our research group spans three institutions with the same firm commitment, as embodied in our institutional declarations and in policies that govern hiring and supervision of staff. As stated by our lead institution (A):

[Our] University is deeply committed to the dignity and equality of all persons—inclusive of sex, gender, marital status, pregnancy, race, color, ethnicity, national origin, age, disability, religion, sexual orientation, gender identity or expression, and veteran status.

Creating and sustaining a positive and inclusive working environment: Our multi-institution group is going on ten years of productive collaboration. Over that time we have mentored 7 postdocs, 4 grad students, and 18 undergrads in research activity and career development. Throughout this collaboration, we have endeavored to recruit a diverse pool of talent from anywhere in the world and to foster a real sense of belonging to all members of the collaboration. Consistent with our clearly stated institutional policies, when we bring members into the group we make active efforts to solicit applications from a diverse pool, to ensure a diverse short list, and to interview in a way that accounts for differences in talents, ideas, and perspectives. These approaches have led to an evolving collaboration that is diverse with respect to multiple axes.

To our group, “inclusion” means “belonging” to a tight knit group of scientists that cultivates a welcoming and positive working environment. The senior members work especially hard to keep the collaboration “flat”—the opposite of hierarchical—with all members making contributions at a working level and avoiding a “boss and employee” mode of work. We actively solicit ideas from all members when discussing scientific findings and in preparing publications. It is not enough to be open to comments from everyone – we must actively *ask* everyone to speak up. Our goal is to create a group in which everyone can carry out their research with encouragement, support, all the necessary resources, and with a feeling that they are part of a team that is greater than the sum of its parts, no matter where they came to us from. That is what “inclusion” and “belonging” mean to us.

Contributions to the training and development of a diverse and inclusive scientific workforce: Mentoring our early career researchers into the next stage of their careers is a critically important goal for our group. It is no good building a diverse and inclusive group in our home institutions if we cannot send them off into the scientific workforce for stable careers. Thus it is essential to provide them with career mentoring in addition to the sense of belonging.

The postdoc to be funded by this program will also participate in a formal program. Each postdoc already has a “supervisor” with whom they work—the person who supports them with grants and manages their day-to-day work—for this program the “supervisors” are the PI and CoIs. In addition, each postdoc is assigned another faculty member as a mentor. By design the mentor is to be relatively independent of the supervisor so that the postdoc has another senior person looking out for their interests both in their career and in our institution. The mentor usually has related science interests but generally is

not a direct collaborator. The mentor provides overall career guidance, acts as a confidential sounding board for issues, and helps with job searches and career development. In addition, the program organizes regular group mentoring and career guidance seminars hosted by the postdoc mentors and open to all local postdocs. Regular events include:

- Proposal writing workshop: how to write a strong science case, build teams, make strong arguments, and anticipate objections. Covers a wide range of ground and space facilities.
- Job application workshop: how to write a cover letter, choose writers of recommendation letters, craft a compelling research statement, and stand out from the crowd.
- Job interview workshop: how to get on the short list, how to present yourself, how to avoid pitfalls, how to pitch a talk to the relevant audience.
- Practice talks: Postdocs on the job market who are shortlisted and will be interviewed give a practice talk before a few staff members and receive intensive, real time feedback.

Finally, this approach effectively teaches postdocs how to foster inclusion and belonging and how to build their own groups in this way when their time comes. By fostering a sense of inclusive belonging in our own group, and encouraging our early career researchers to carry these values into their own groups, we are doing our best to contribute to the training and development of a diverse and inclusive scientific workforce over the long term.

As with the postdoctoral researcher at Institution A, the undergraduate student funded by this program at Institution B will participate in extensive mentoring and professional development. Institution B has a “laddered” mentoring program where participants at a given career stage have a professional mentor (explicitly not their immediate supervisor) who they regularly engage with about relevant issues including job searches, preparing application materials, choosing research projects, mentoring, etc. As part of this laddered program, they also mentor those at an earlier stage. In addition, CoI B4 has adopted many of the best practices advocated for by University of Wisconsin’s Center for the Improvement of Mentored Experiences in Research (CIMER; <https://cimerproject.org/>), including structured annual evaluations and efforts to increase collaboration and peer mentoring within the research group (the latter of which is explicitly intended to increase mutual support among group members, which facilitates retention and a sense of belonging). This mentoring effort also leverages Institution B’s extensive professional development opportunities, which include a variety of workshops in proposal-writing, job searches and interviews, teaching techniques, mentoring, and so on. These efforts help to prepare trainees for their next career stage and to encourage positive work habits both at Institution B and in the future, and we hope that they contribute to the support and creation of a diverse scientific workforce.

10. Anonymized Table of Personnel and Work Effort

Inst.	Label	Role	Responsibilities	Effort	Funded
A	PI, A1	Faculty	oversee and manage entire program mentor postdoc (A2) assist with simulations and analysis	20%	0%
A	Co-I, A2	Postdoc	implement/optimize new code items run primary simulations lead dwarf environment analysis	100%	100%
A	Co-I, A3	Faculty	mentor postdoc (A2) analysis from the parent big box assist with simulations and analysis	20–25%	0–25%
B	Co-I, B4	Faculty	supervise undergrad student implement and deliver code improvements assist with simulations and analysis	16.7%	8.3%
B	UG	Undergrad Student	test new code items lead chemical enrichment analyses	42%	42%

Table 1 : Table of Work Effort by the Co-Is and supported personnel, broken down by institution. Levels of Effort are given as “percent of full-time equivalent effort per year” such that a person working 100% of their time on this program for three years is listed at “100%.” Due to the nature of the funding for some investigators’ positions, the level of committed effort (“Effort”) is larger than the level of effort funded by this proposal (“Funded”).

11. Redacted and Anonymized Budgets and Narratives

BUDGET NARRATIVE JUSTIFICATION

Proposal Title: Galaxy Formation at the Edge: Resolving the Smallest Galaxies in the Local Universe

Sponsor: NASA – Astrophysics Theory

Period of Performance: 02/01/2024-08/31/2027

Salaries – Institution A

PI A1 and Co-I A3 – The PI and Co-I A3 request no funds to cover their salary as they have 12-month salaries.

Co-I A2 – This budget requests funds to cover 12-person months (100% effort) of Co-I A2 as a postdoctoral fellow for three years. Co-I A2 will take on the following major work items: (1) participate in the selection of galaxies for simulation, (2) lead implementation and testing of some new code elements and participate in those new items led by Institution B, (3) help create, manage, and curate our new simulation runs, (4) lead the analysis of the “external halo”, “edge”, and “field” galaxy simulation scenarios, generating customized code along the way to drive new analyses, in collaboration with the other team members (5) create and maintain code in an open source repository, contributing modules back to community based codes (eg. *yt* and *powderday*) as appropriate, (6) collaborate with the Co-I #A3 to adapt and expand our mock data pipeline to generate the proposed products, (7) support the release of those products in open repositories, and (8) lead and/or participate in the conceptualization, composition, editing, and publication of all group papers.

Annual salary increases of 5% are included in the calculations.

Other Direct Costs

Escalation: An annual increase of 3% is included in the calculations for other direct costs, except for graduate student tuition and insurance fees, for which annual costs are provided.

Materials/Supplies:

We request \$4,000 for a laptop to be used by Co-I A2 to develop, run, and analyze data.

We request \$1,673 for DropBox subscriptions for the PI and Co-Is and the undergraduate students. This will be used for sharing collaboration documents. The current cost of a “DropBox Plus” subscription (2 TB of storage) is \$10/month plus sales tax (so roughly \$125 total per subscription per year; we request 3 subscriptions).

The total requested for materials and supplies is \$5,673.

Publications:

We estimate we will publish one paper per year in The Astrophysical Journal to disseminate the results of the funded research. The cost has been calculated using the digital submission (digital quanta) costs charged by IOP Publishing and historical costs for similar manuscripts, based on 100 quanta per paper, \$4,800 is estimated for year 2, \$4,900 is estimated for year 3, and \$5,000 is estimated for year 4.

The total requested for publications is \$14,700.

Subawards: Institution B

A subaward is requested for Institution B. They will work with the overall PI to establish and track milestones, to supervise the reduction and analysis of data from the Enzo simulations and for comparisons with observations, as well as for participating in the writing of manuscripts for publications. The bulk of their time will be devoted to the development, debugging, and maintenance of new software modules for Enzo in support of the scientific deliverables of the grant. The detailed budget for the subaward is attached.

Travel

Domestic Travel:

Direct face-to-face collaboration is essential at some stages of the project, such when production simulations are being planned and when major scientific results are being interpreted and debated prior to being written up. In the case of our program, for which we are adding new code elements to Enzo, direct collaboration is the most efficient means of the team who implemented a given feature to train the others on its use. As Co-I #B1 will visit Institution A several times, we have therefore budget only collaboration trip for Co-I #A2 to visit Institution B during the critical Year 1 of the program.

Year 1: Collaboration visit to Institution B: We plan for one visit for the Co-I A2 to work with Co-I B1. The trip is planned for one week with lodging estimated at \$763 with the daily rate at \$109/day. Meals and incidental rates are estimated at \$462 with the daily rate at \$66/day. Round-trip local transportation to/from the airport is estimated at \$39. Airport parking is estimated at \$58. local travel at the destination is estimated at \$200. Air tickets are estimated at \$390. The trip is budgeted for one traveler, 7 days, for a total cost of \$1,912.

International Travel:

Funding is requested for Co-I #A2 to attend international conferences for years 2 and 3. Since the exact conferences are not known yet, we use Potsdam, Germany for year 2 budget estimates and Canberra, Australia for year 3 budget estimates.

Year 2: Potsdam, Germany (2025): The lodging is estimated at \$1,110 with the daily rate at \$222/day. Meals and incidental rates are estimated at \$660 with the daily rate at \$132/day. Round-trip local transportation to/from the airport is estimated at \$92. Airport parking is estimated at \$74. Registration is estimated at \$600. local travel at the destination is estimated at \$300. Air tickets are estimated at \$1,923. The trip is budgeted for one traveler, 5 days, for a total cost of \$4,759.

Year 3: Canberra Australia (2026): The lodging is estimated at \$1,040 with the daily rate at \$208/day. Meals and incidental rates are estimated at \$590 with the daily rate at \$118/day. Round-trip local transportation to/from the airport is estimated at \$39. Airport parking is estimated at \$44. Registration is estimated at \$618. local travel at the destination is estimated at \$309. Air tickets are estimated at \$3,000. The trip is budgeted for one traveler, 5 days, for a total cost of \$5,640.

The total requested for travel is \$12,311.

Galaxy Evolution at the Edge

Institution A		Project Dates:				Start	End									
						2/1/2024	8/31/2027									
Title:		Galaxy Formation at the Edge: Resolving the Smallest Galaxies in the Local Universe				2/1/2024 to 1/31/2025		2/1/2025 to 1/31/2026		2/1/2026 to 1/31/2027		2/1/2027 to 8/31/2027				
Expense Category	Description	Role/Title	Pro-rated Base	% Effort	Person Mos	Salary	% Effort	Person Mos	Salary	% Effort	Person Mos	Salary	% Effort	Person Mos	Salary	Total
Other Direct Costs	Materials/Supplies/Chemicals					\$4,400			\$412			\$424			\$437	\$5,673
	Publications					\$0			\$4,800			\$4,900			\$5,000	\$14,700
	Quanta per year	-	100	100	100	-										\$0
	Price per quanta	\$47	\$48	\$49	\$50											\$0
	Total Publication Cost	\$0	\$4,800	\$4,900	\$5,000											\$0
	Consultant Services					\$0			\$0			\$0				\$0
	Data Management Plan Costs					\$0			\$0			\$0				\$0
	Computer Services					\$0			\$0			\$0				\$0
	# of machines requiring data backup	-	-	-	-	-										\$0
	Annual Data Backup Rate															\$0
	Total Computer Services		\$0	\$0	\$0	\$0										\$0
	Subcontract-F&A Bearing up to \$25K					\$25,000										\$25,000
	Subcontract-non F&A Bearing > \$25K	Institution B				\$42,739			\$70,232			\$78,261			\$69,834	\$261,066
	Travel-Domestic (from travel calculator)					\$1,912						\$5,640				\$1,912
	Travel-Foreign (from travel calculator)					\$0			\$4,759							\$10,399
Other						\$0			\$0						\$0	
Equipment > \$4,999						\$0										\$0
Buffer																
Total Other Direct Costs						\$74,951			\$80,203			\$89,225			\$75,271	\$318,750
Total Direct Costs						\$165,478			\$176,201			\$190,022			\$75,271	\$606,972

Planning Tool for trips				
Domestic Travel				
Trips				
Destination	MI	May 2024		
Meals & Incide Rate		60.00		
Mileage Rate		\$0.655		
# of miles		60		
Accommodation Rate		\$109.00		
# of days		7		
Meals & Incidentals		462.00		
Mileage		39.00		
Accommodation		763.00		
Registration		-		
Parking at BWI		56.00		
Ground Transportation at Destination, etc.		200.00		
Airfare		390.00	-	-
Other		-	-	-
Cost per trip		1,912.00	-	-
# of Travelers		1	1	1
Period		1	1	2
Total per trip		1,912.00	-	-

Foreign Travel				
Trips				
Destination	2025	2026		
Potsdam, Germany		Caliberna, Australia		
Meals & Incide Rate	132.00	118.00		
Mileage Rate	\$0.655	\$0.655		
# of miles	140	60		
Accommodation Rate	\$222.000	\$208.000		
# of days	5	5		
Meals & Incidentals	660.00	\$99.00		
Mileage	92.00	39.00		
Accommodation	1,110.00	1,040.00		
Registration	600.00	618.00		
Membership	-	-		
Ground Transportation at Destination, etc.	300.00	309.00		
Parking Airport at IAD	74.00			
Parking Airport at BWI		44.00		
Airfare	1,923.00	3,000.00		
Other		-		
Cost per trip	4,759.00	5,640.00		
# of Travelers	1	1		
Period	2	3		
Total per trip	4,759.00	5,640.00		

Budget narrative-Institution B

The subaward PI will have primary responsibility for managing the institution's portion of the grant budget, for working with the overall PI to establish and track milestones, to supervise the reduction and analysis of data from the Enzo-E simulations and for comparisons with observations, as well as for participating in the writing of manuscripts for publications. The bulk of the subaward PI's time will be devoted to the development, debugging, and maintenance of new software modules for Enzo in support of the scientific deliverables of the grant.

Faculty summer salary: In Years 1-4 of the grant, we request partial summer salary support (1 month at 100% effort) to fund the subaward PI. In total, this individual will commit 16.7% of their time (2 months) per calendar year to this project.

Undergraduate student support: We request funding for an undergraduate astrophysics student to work on this project for its duration (and we acknowledge that the student in question may be two students who are employed sequentially). The student(s) in question will work primarily on analysis of the simulations created as part of the project. This student will start working in Summer 2024 and continue through Summer 2027. We assume that they will work 10 hours/week during the academic year (14 weeks/semester) and then full time during the summers (40 hours/week for 14 weeks). This student will spend 100% of their funded time on this project.

Travel: We request \$6K of support per year for domestic travel for all four years of this project. Two trips are for the subaward PI to visit the lead institution, and the third trip is for either the subaward PI or the undergraduate student to attend a domestic conference to present results relating to this project.

The budget for each collaboration visit (2 total/year) is estimated to be roughly \$1,500, broken down as follows: a rental car @ \$150, hotel \$600 (4 nights @ \$150/night), plane ticket @ \$350, per diem of \$300, local ground travel including parking and mileage at \$100 total.

Based on recent conference travel, the budget for each conference trip (1 total/year) is estimated to be \$3,000, broken down as follows: hotel \$1,500 (6 nights @ \$250/night), conference registration fee \$400, plane ticket \$600, per diem \$300 total., local ground travel including parking and mileage at \$200 total.

We furthermore request funding in Year 3 for an international conference for the subaward PI or the undergraduate student. Based on recent conference travel, the budget for an international conference trip is estimated to be \$3,500, broken down as follows: hotel \$1,200 (6 nights @ \$200/night), conference registration fee \$400, plane ticket \$1,400, per diem \$300 total., local ground travel including parking and mileage at \$200 total.

Materials and supplies: We request \$400/year for materials and supplies for all three years of this research project. This is broken down as follows:

\$250/year for DropBox subscriptions for the subaward PI and the undergraduate students. This will be used for sharing collaboration documents. The current cost of a “DropBox Plus” subscription (2 TB of storage) is \$10/month plus sales tax (so roughly \$125 total per subscription; we request 2 subscriptions).

We request an additional \$150/year for additional miscellaneous supplies such as data storage devices (an 8-TB external hard drive is approximately \$150 on newegg.com; a 256 GB USB memory stick is \$37 on newegg.com), poster printing (\$6 per linear foot, ~\$25 per poster), textbooks on subjects related to the grant (typically \$100/book), and other similar charges.

NASA Anonymized Budget								
	Year 1		Year 2		Year 3		Year 4	Total
<u>Equipment</u>	-		-		-		-	-
Total Equipment	-		-		-		-	-
<u>Travel</u>								
Domestic	6,000		6,000		6,000		6,000	24,000
Foreign	-		-		3,500		-	3,500
Total Travel	6,000		6,000		9,500		6,000	21,500
<u>Participant Support Costs</u>								-
Total PSC	-		-		-		-	-
<u>Other Direct Costs</u>								
Materials & Supplies	400		400		400		400	1,600
Publication/Dissemination	-		-		-		-	-
Consultant Services	-		-		-		-	-
Computer Services	-		-		-		-	-
Subawards	-		-		-		-	-
Other	-		-		-		-	-
Tuition & Fees	-		-		-		-	-
Total Other Direct Costs	43,146		44,734		49,848		44,480	182,208